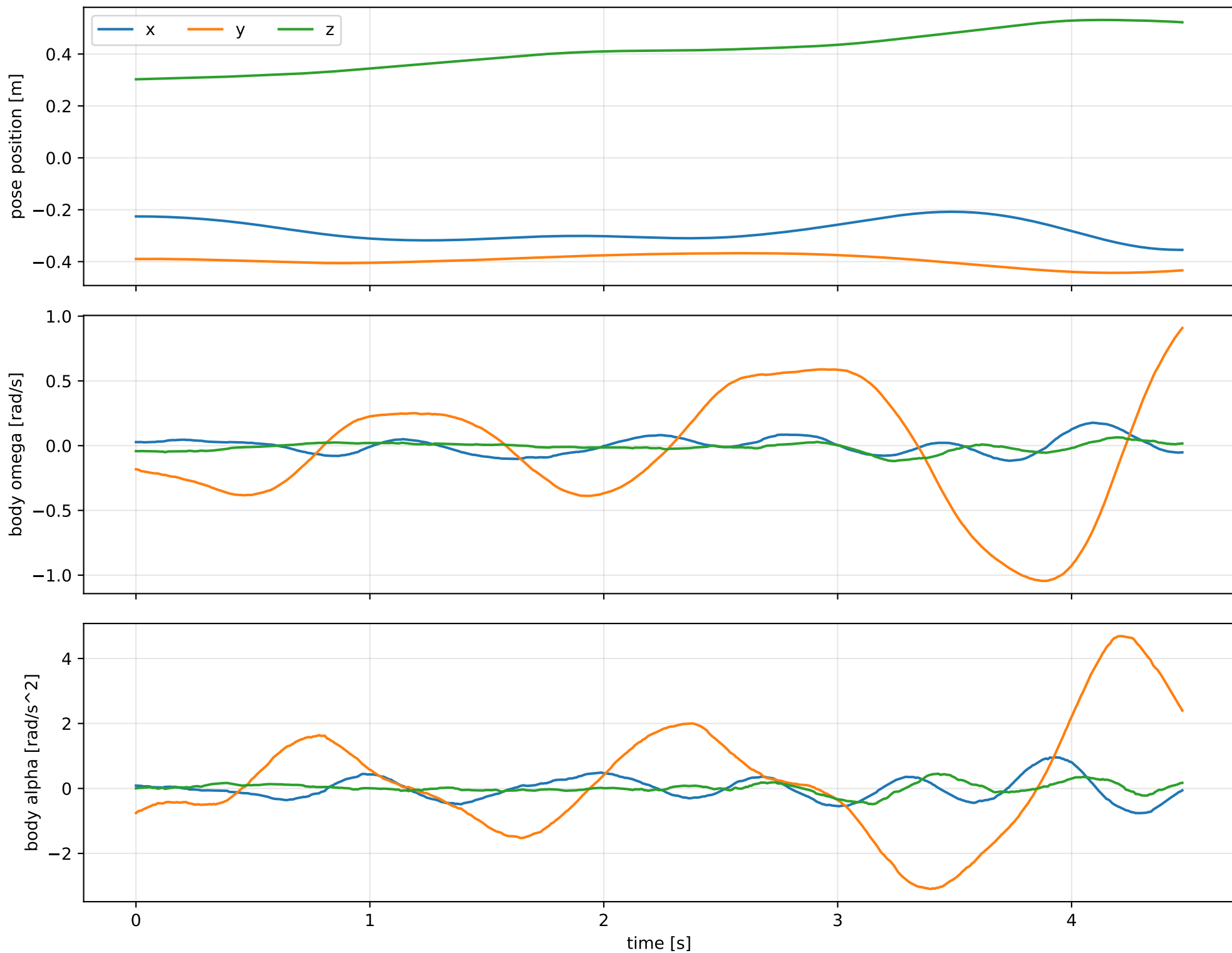
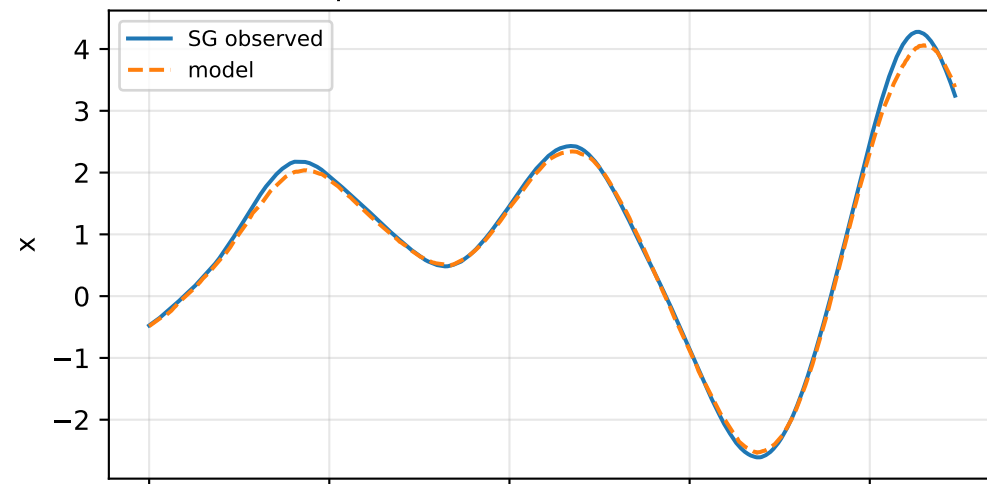


solver_standard_gauge_least_squares: SG trajectory and derived motion

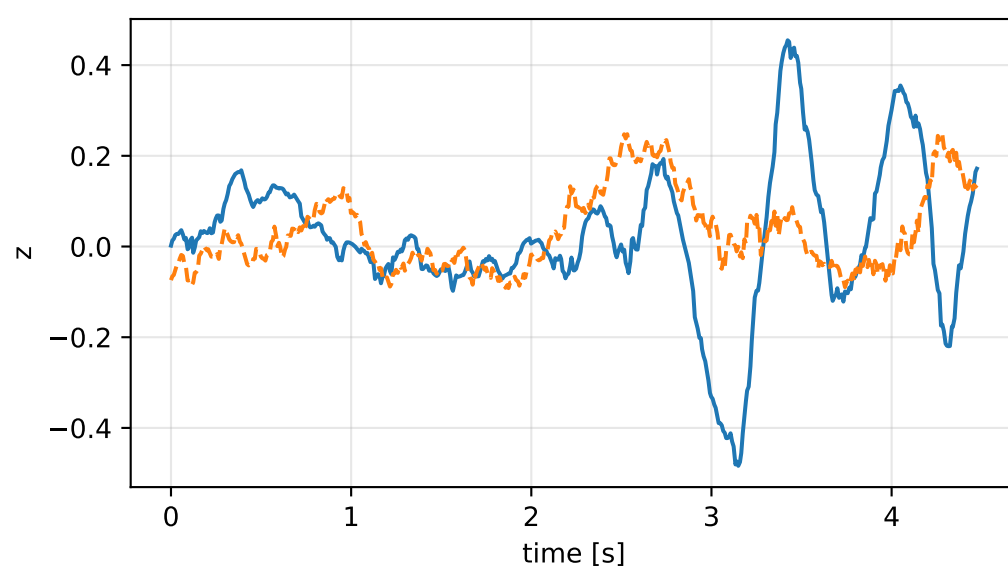
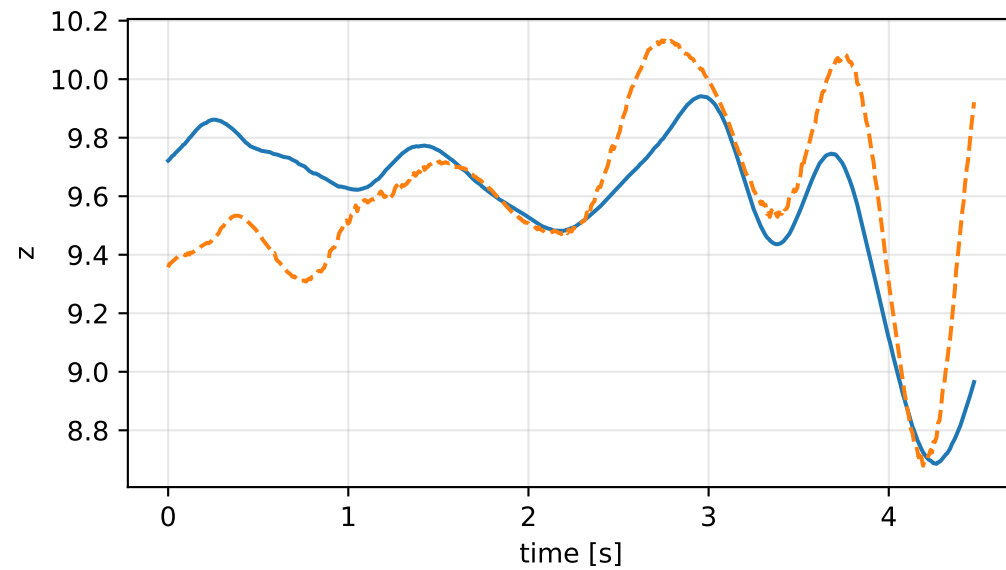
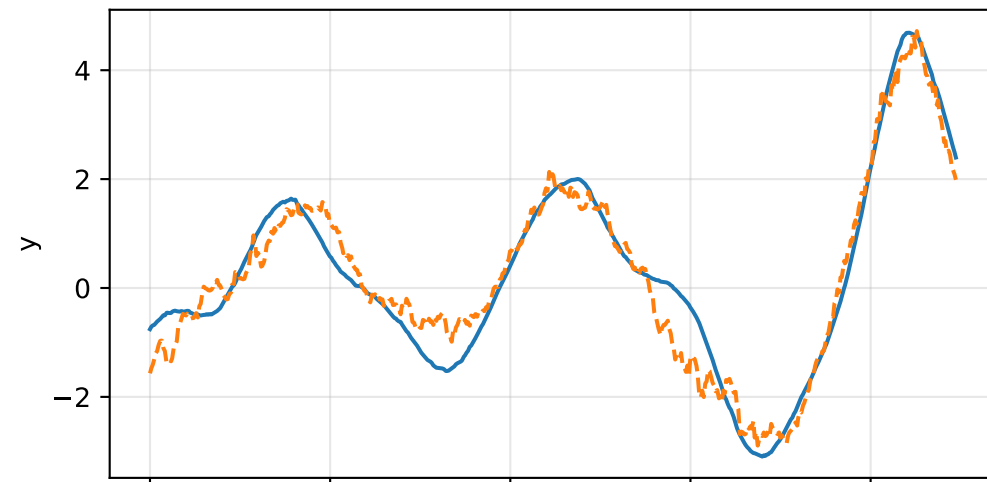
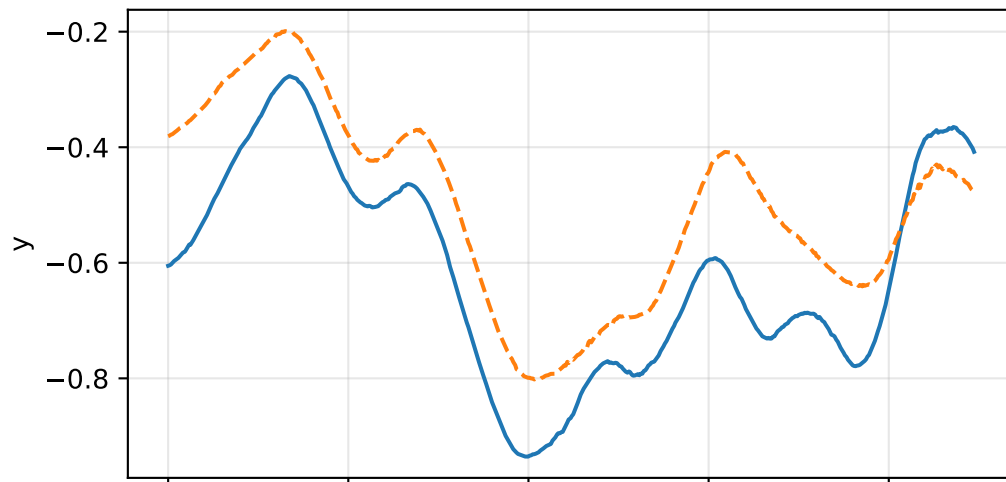
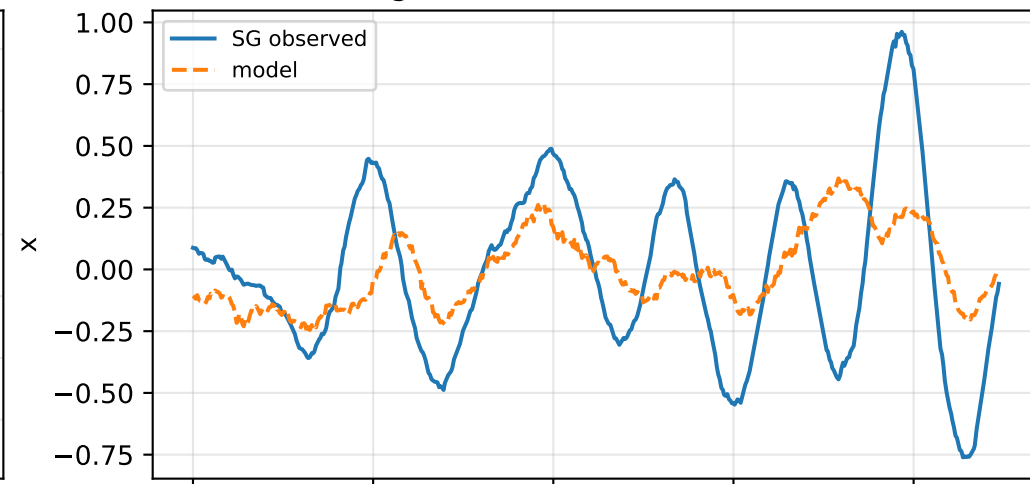


solver_standard_gauge_least_squares: acceleration residual objective

specific acceleration [m/s²]

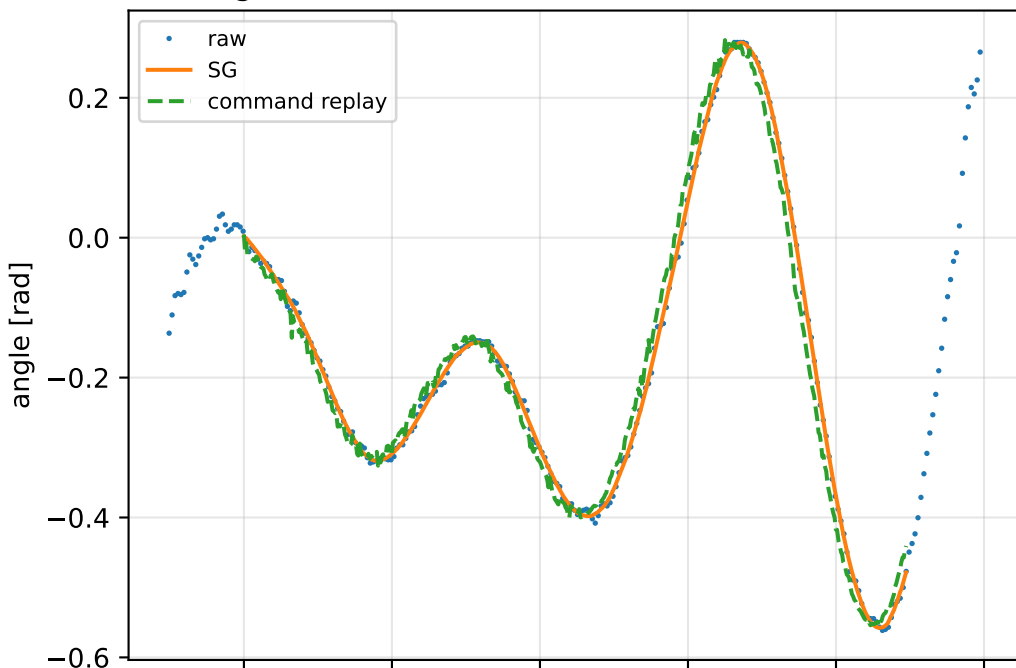


angular acceleration [rad/s²]

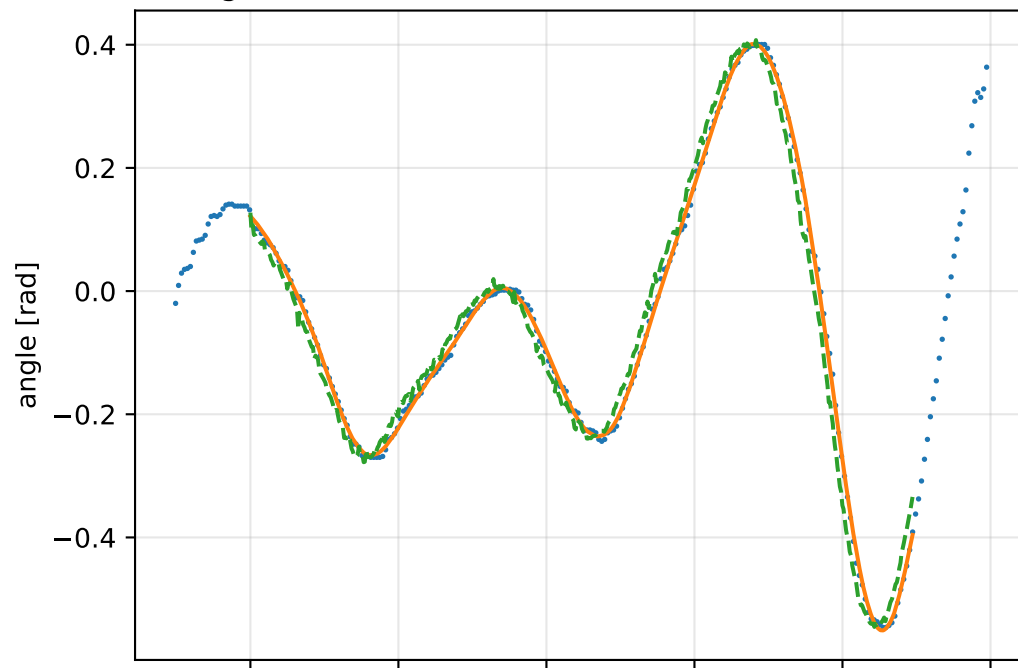


solver_standard_gauge_least_squares: gimbal measurement audit (objective=measured_sg)

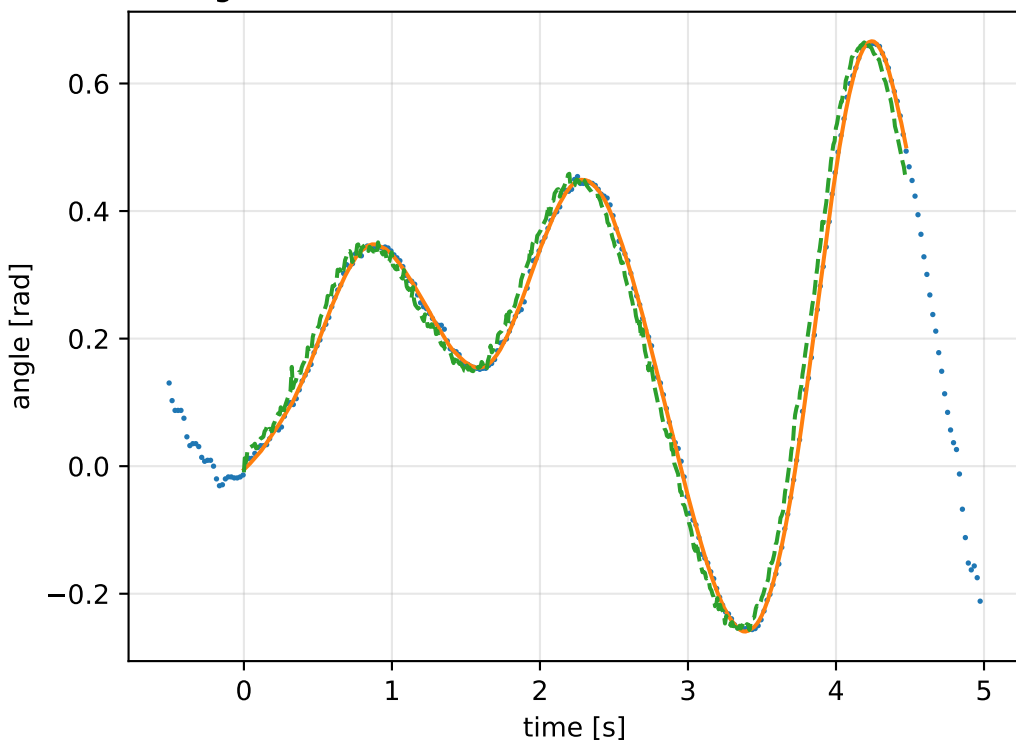
gimbal 1: RMSE=0.027, max=0.07529 rad



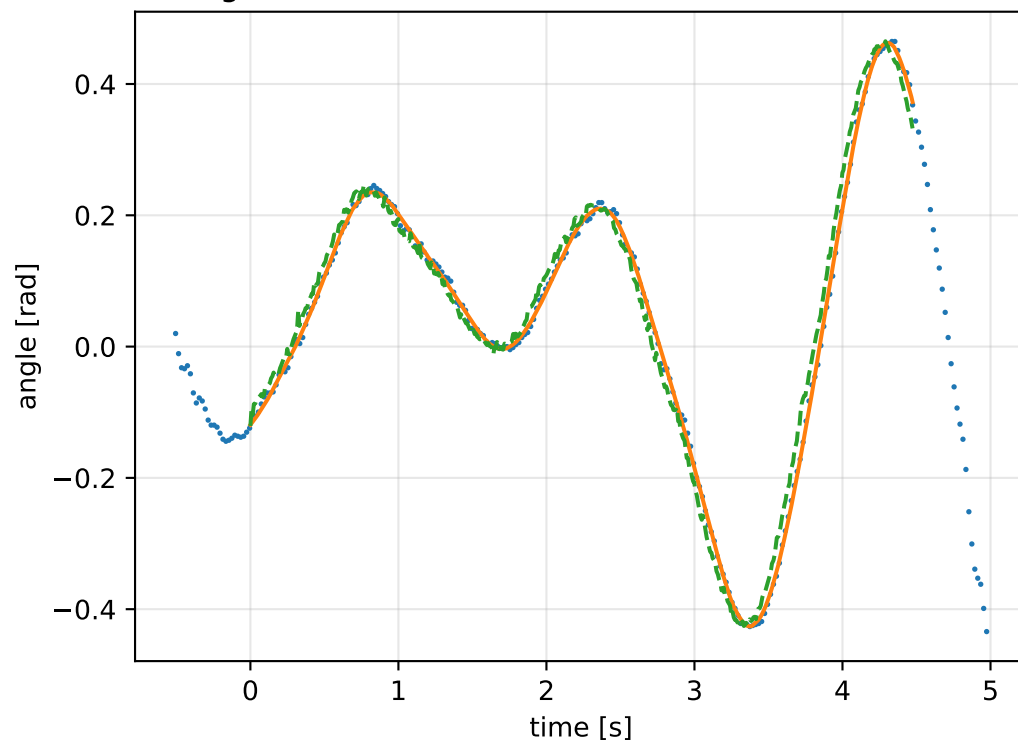
gimbal 2: RMSE=0.03107, max=0.08239 rad



gimbal 3: RMSE=0.03106, max=0.08682 rad

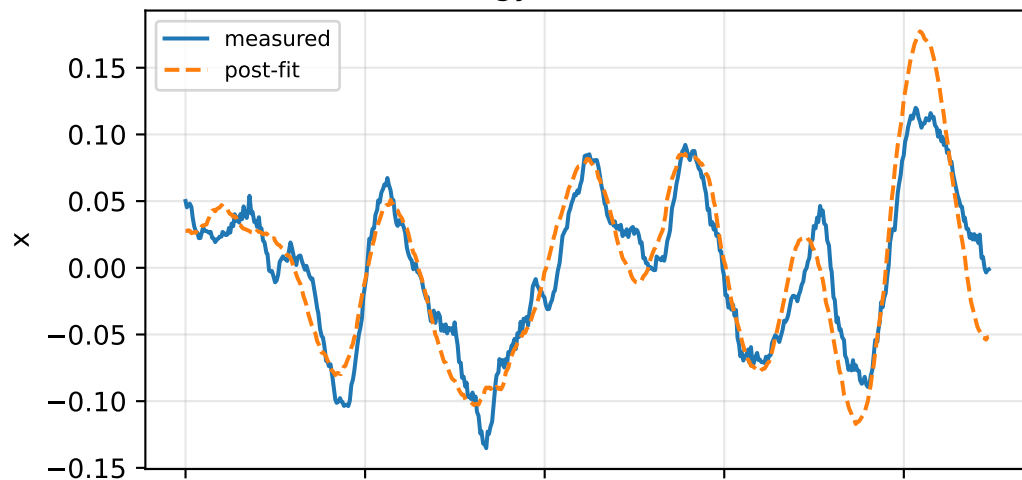


gimbal 4: RMSE=0.02598, max=0.07261 rad

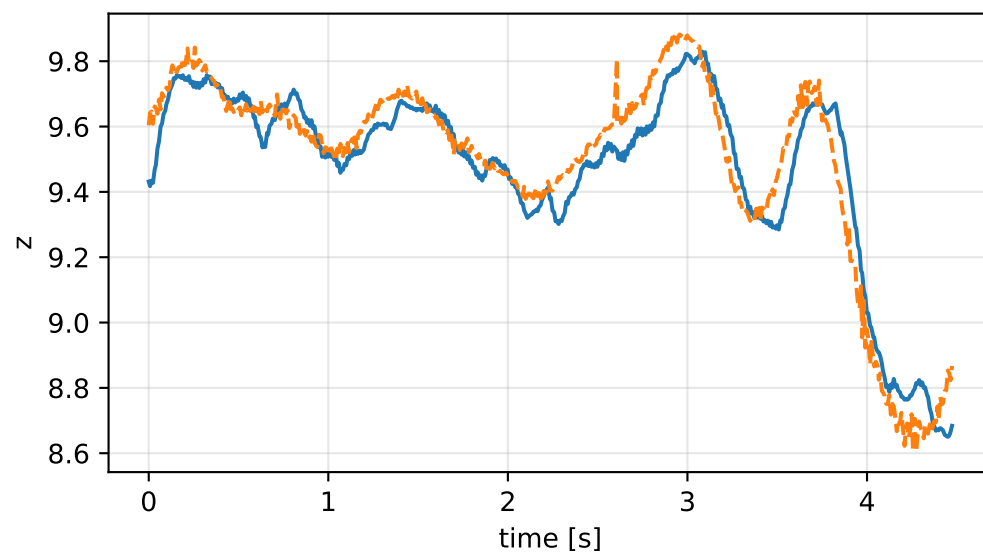
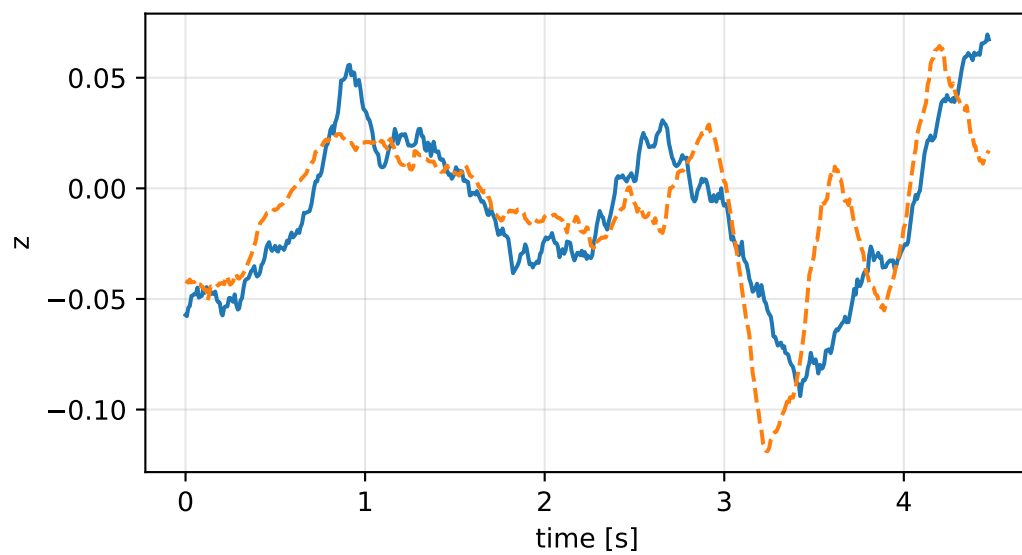
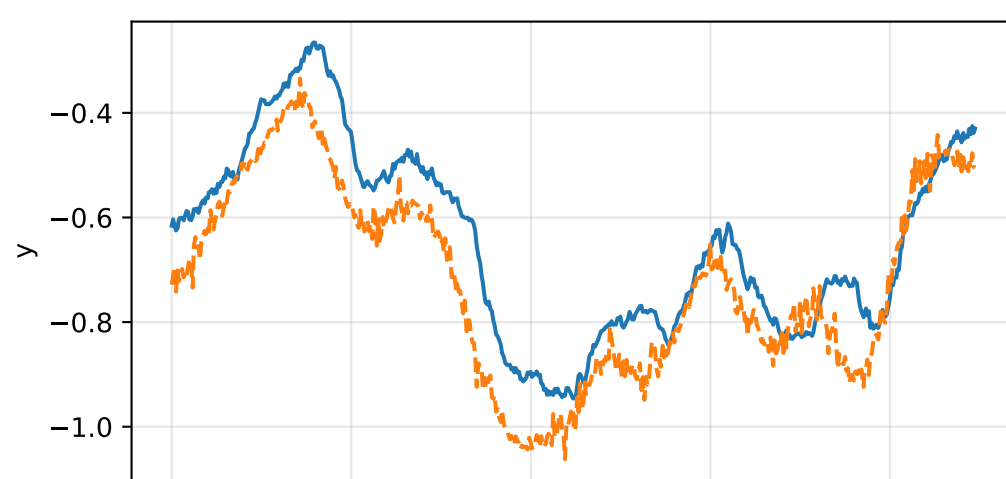
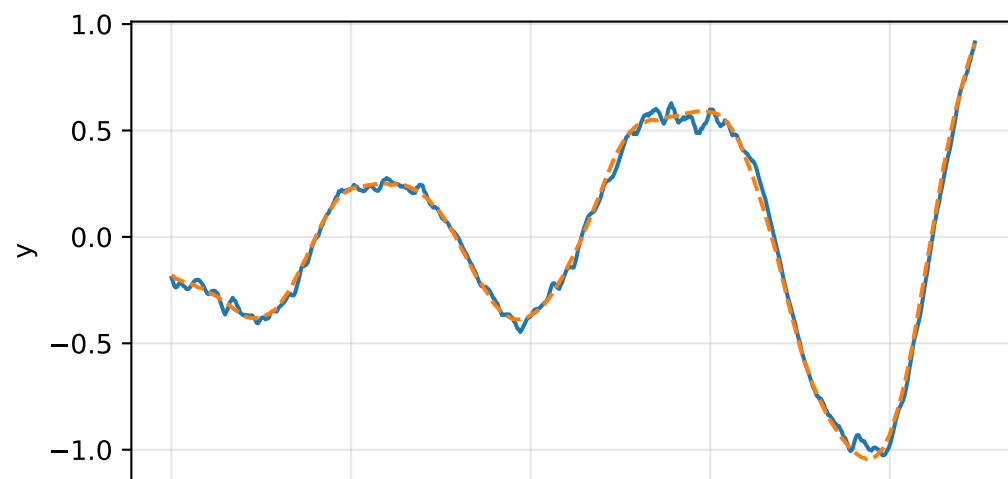
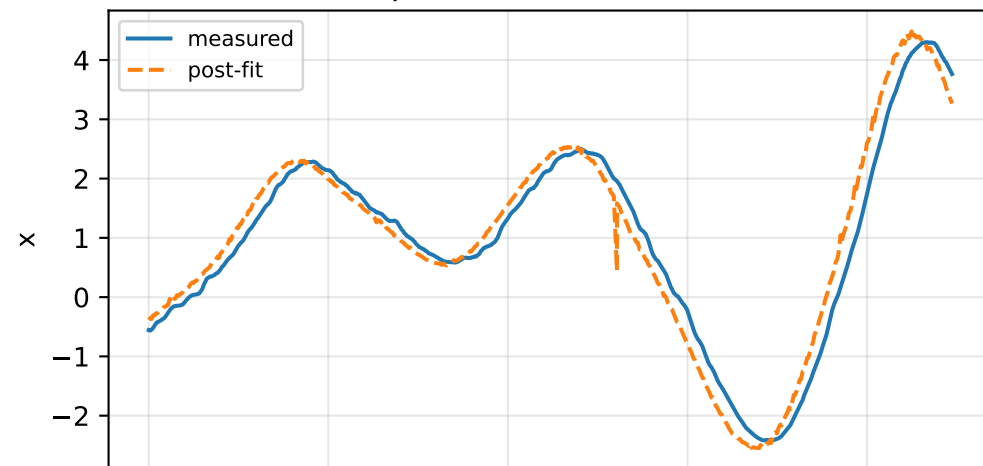


solver_standard_gauge_least_squares: IMU comparison (diagnostic only)

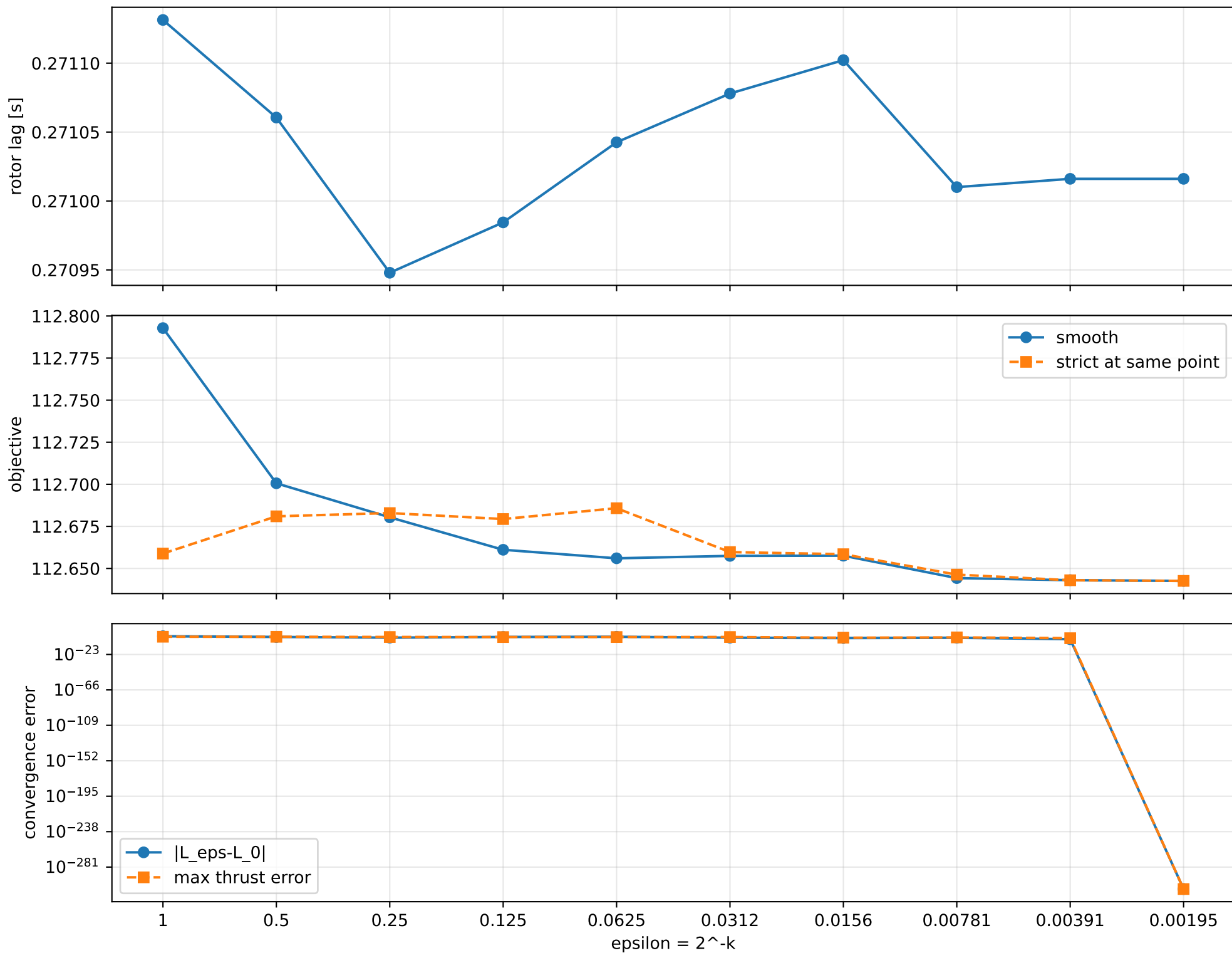
gyro [rad/s]



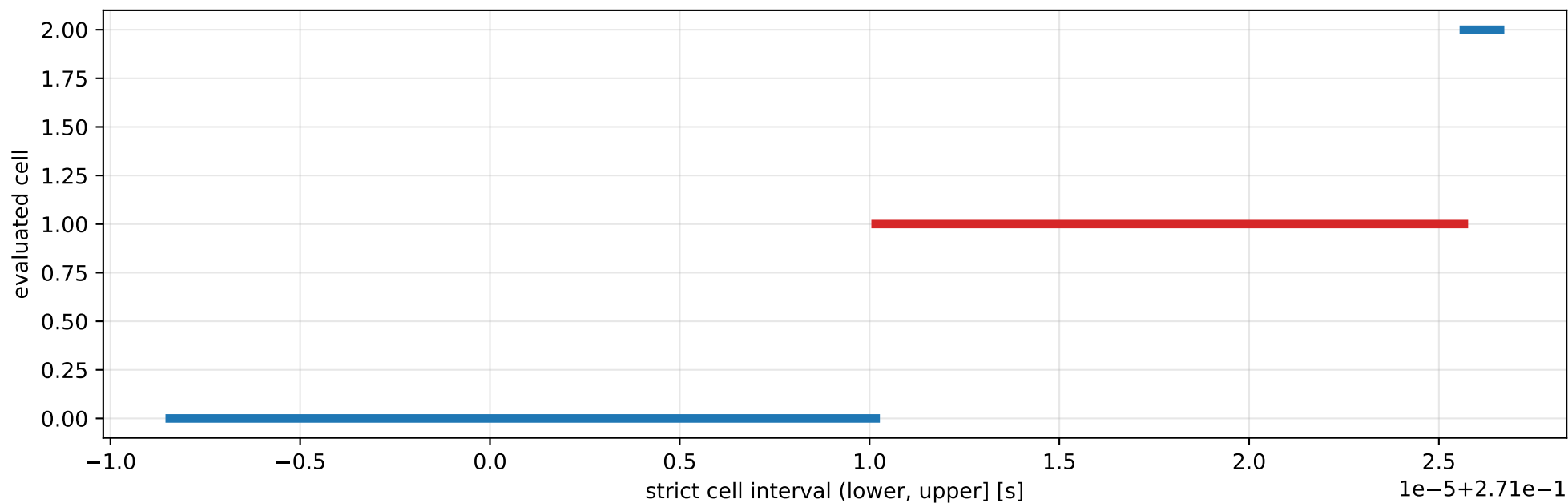
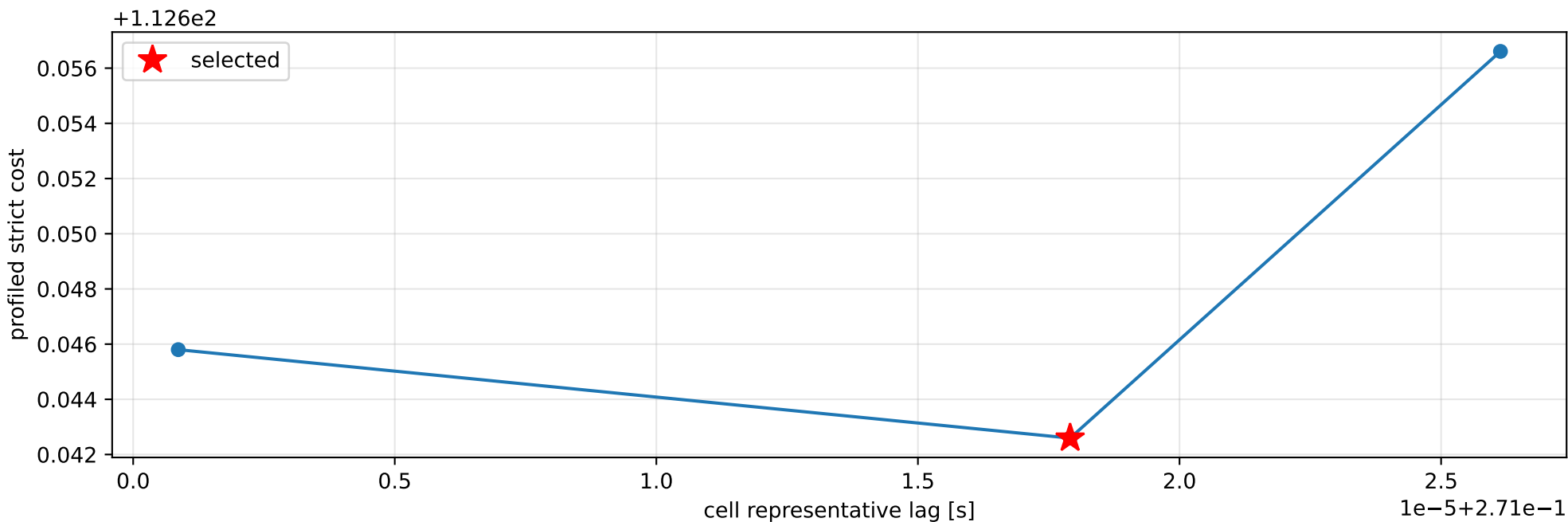
specific force [m/s^2]



solver_standard_gauge_least_squares: smooth-to-strict continuation



solver_standard_gauge_least_squares: exact strict-ZOH lag cells



selected (0.27101016, 0.271025658] s; final smooth 0.271016075 s; data boundary=False

solver_standard_gauge_least_squares: parameter estimate

Case: solver_standard_gauge_least_squares
status: completed (all stages completed)

mass [kg]: nominal 2.35159759; estimated 38910.4095

inertia [kg m²]:

```
[[2869.09379451  71.04793627 -10.82210945]
 [ 71.04793627 1353.59458165 -423.75588231]
 [-10.82210945 -423.75588231 4093.9182161 ]]
```

CoG body [m]: [-0.02644521 0.01135027 0.00940276]

force effectiveness: [14856.04499334 12022.23348132 13263.16180324 15933.23068426]

scale-free J/m [m²]:

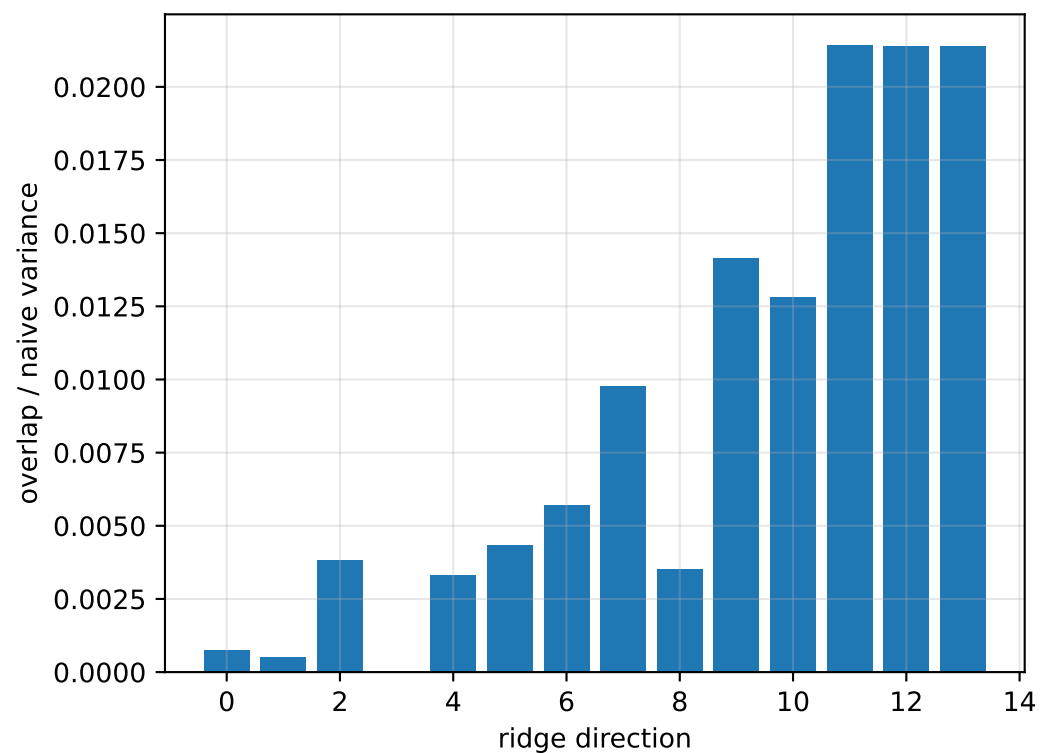
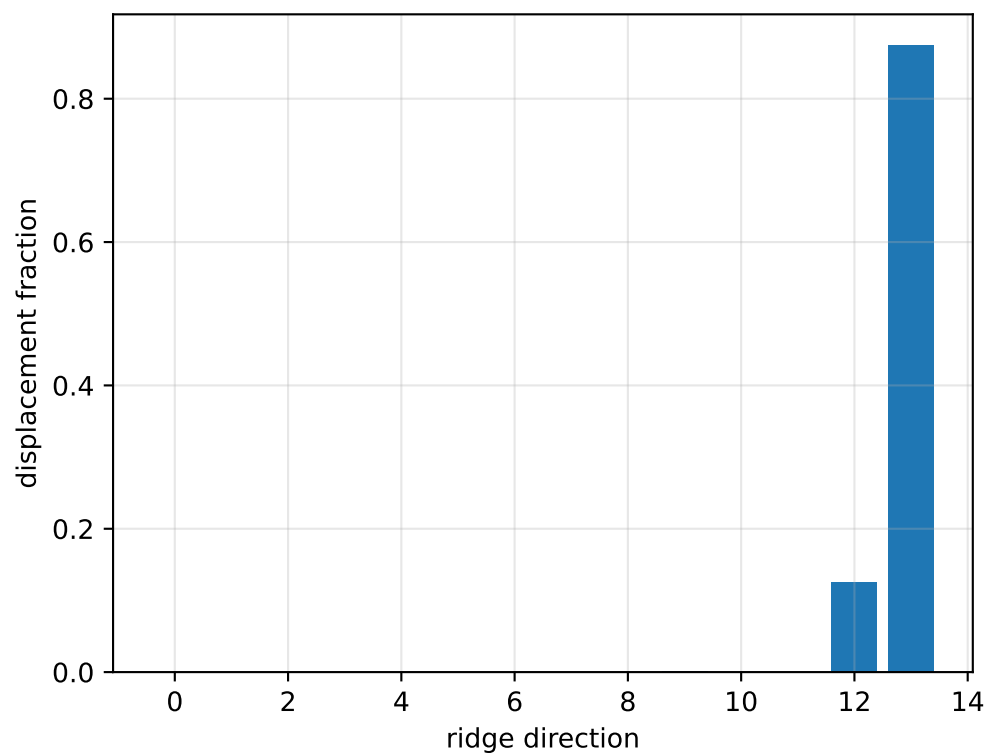
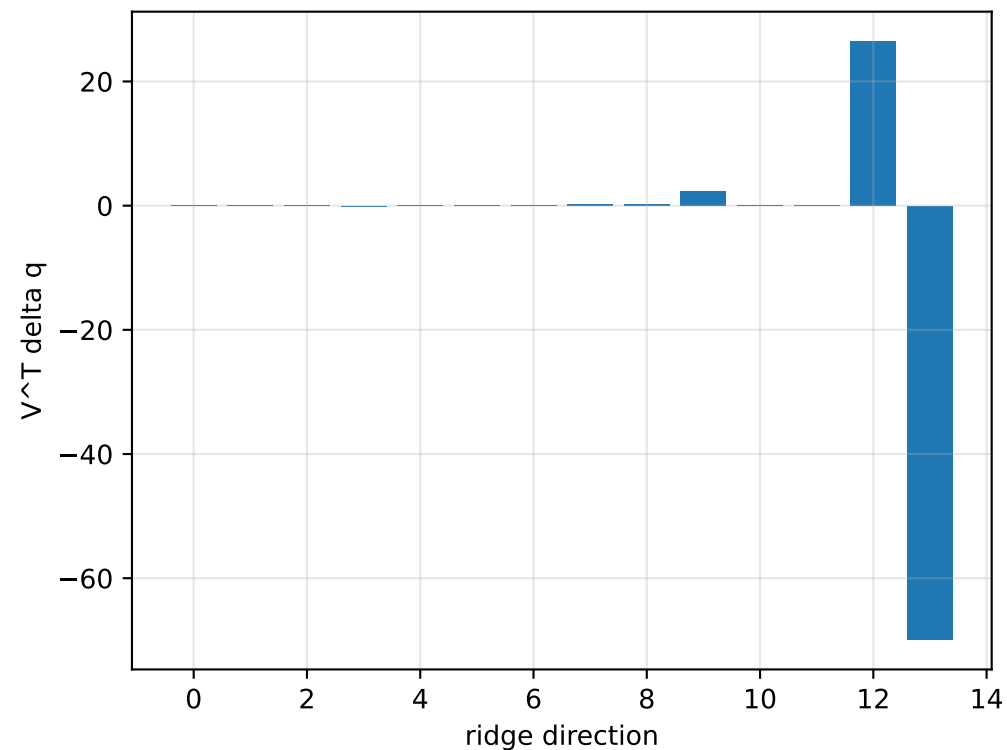
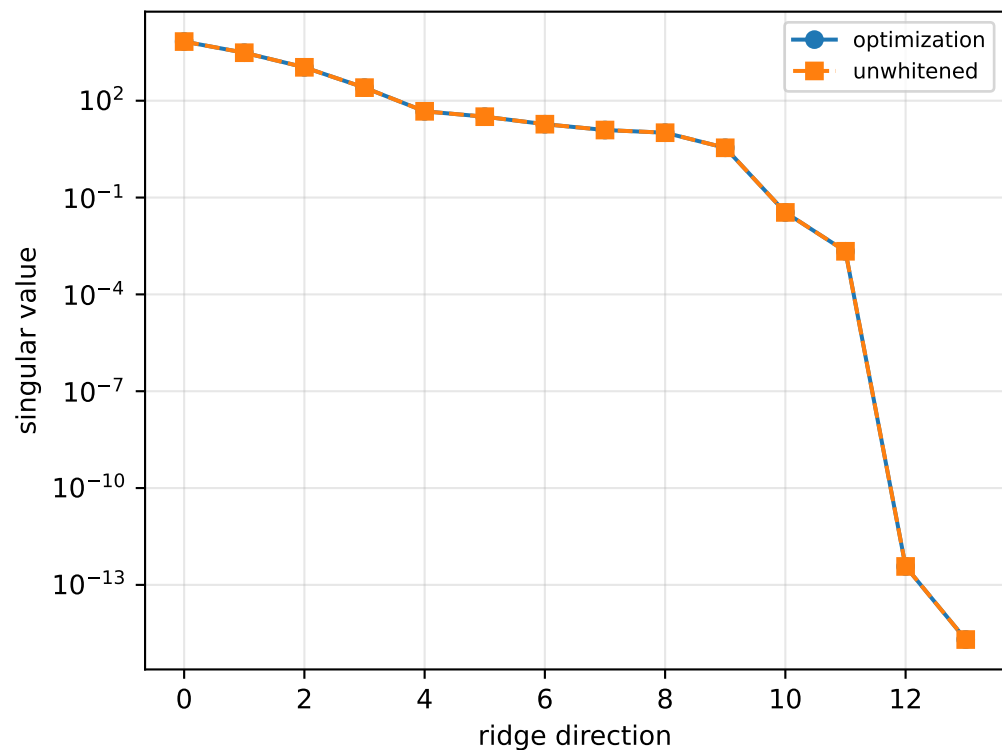
```
[[ 0.07373589  0.00182594 -0.00027813]
 [ 0.00182594  0.03478747 -0.01089055]
 [-0.00027813 -0.01089055  0.10521396]]
```

scale-free f/m: [0.3818013 0.30897217 0.3408641 0.40948504]

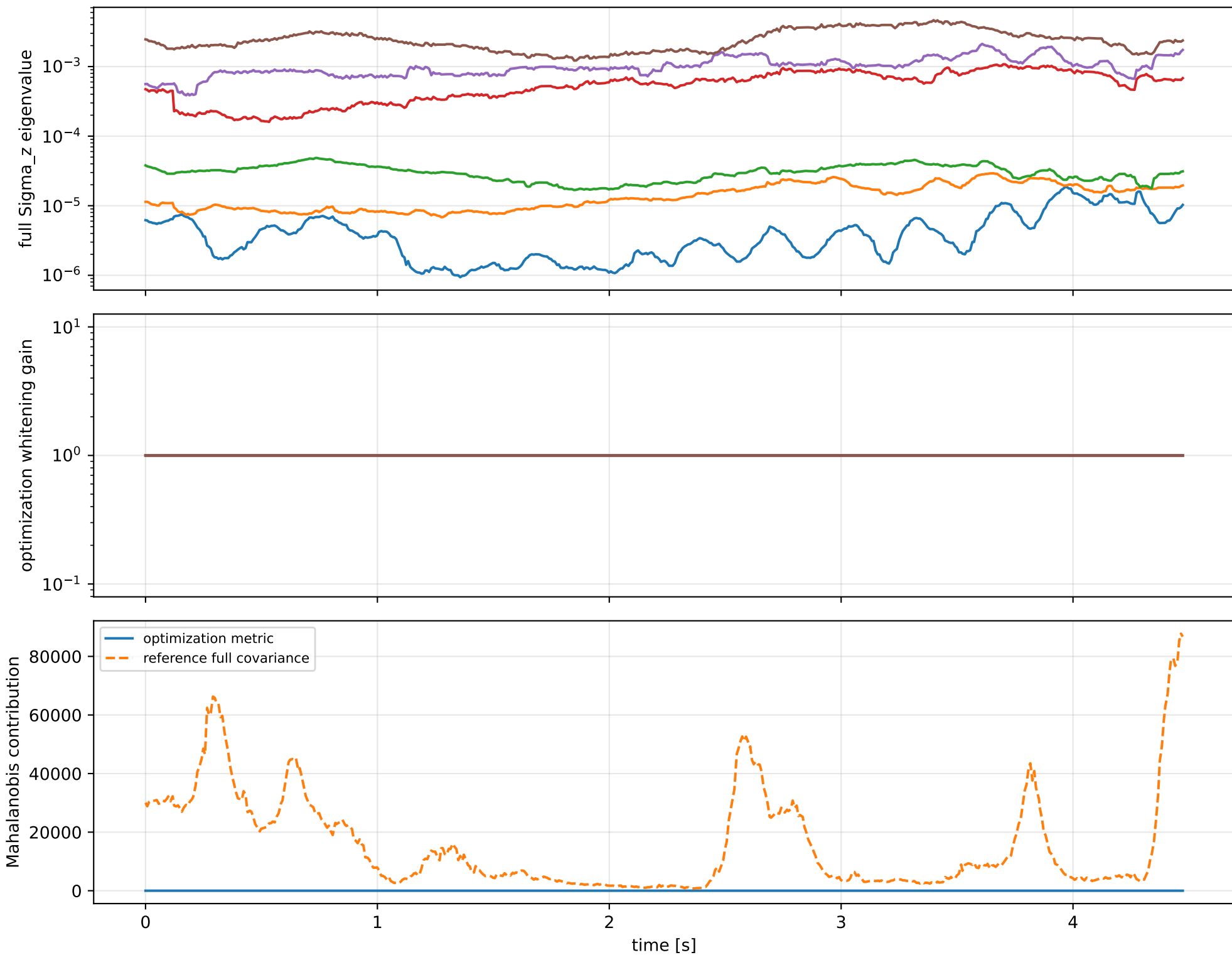
rotor lag cell: (0.27101016, 0.271025658] s

representative: 0.271017909 s

solver_standard_gauge_least_squares: ridge spectrum (rank 12, nullity 2, ||jv||=2.33e-13)



solver_standard_gauge_least_squares: optimization vs reference covariance



solver_standard_gauge_least_squares: wrench / closure (force max 1.02e-10, torque max 1.48e-12)

